

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

11 September 2017

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	25
2	25
Section B	
3 OR 4	25
Total	75

This document consists of **18** printed pages.



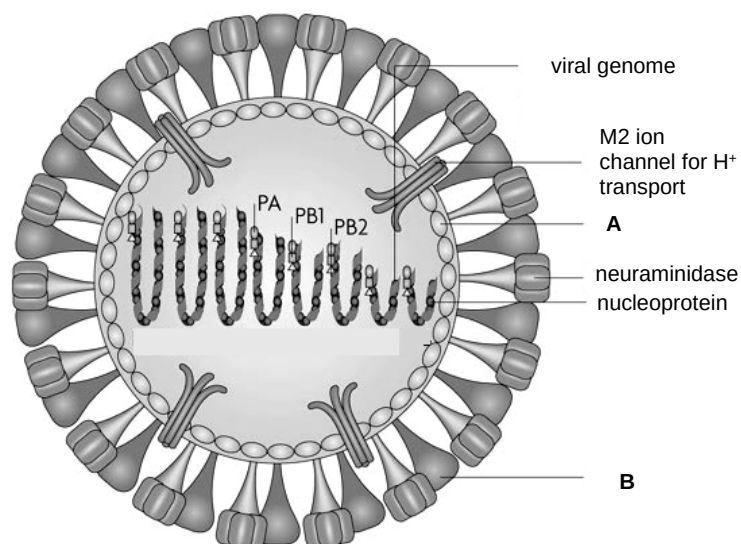
Innova Junior College

[Turn over

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows the influenza virus.



Legend

PA, PB1, PB2 are RNA polymerases

Fig. 1.1

- (a) (i) Identify structures A and B.

A

B

[2]

- (ii) Explain why the influenza virus needs its own RNA polymerase.

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 [2]

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Use

- [5]

- [3]

- [2]

(iii) Explain why yearly vaccination is recommended for influenza.

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Doctors sometimes prescribe antibiotics to patients who are infected by influenza to combat secondary bacterial infections. Gramicidin A is an example of an antibiotic. Fig. 1.2 shows the molecular structure of Gramicidin A.

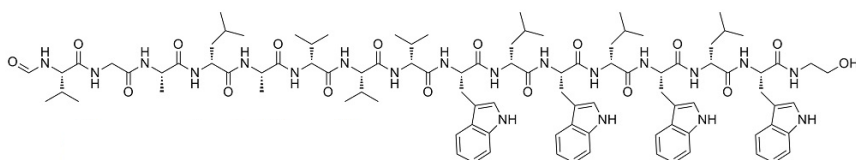


Fig. 1.2

(d) Illustrate the reaction that forms the peptide bonds between two amino acids.

[2]

Gramicidin A folds into a 3-dimensional configuration that inserts itself into the bacterium's cell surface membrane. It allows non-specific movement of ions which eventually cause the bacterial cell to die. Fig. 1.3 shows the interaction of Gramicidin A with the bacterium's cell surface membrane.

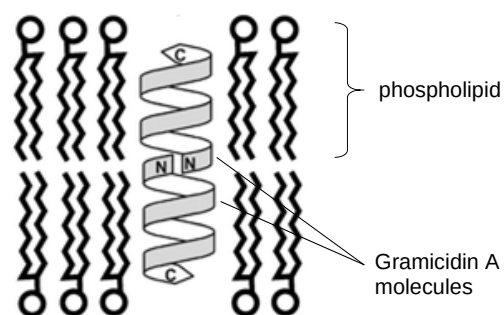


Fig. 1.3

- (e) (i) Describe how the Gramicidin A shown in Fig. 1.2 folds into the 3-dimensional structure shown in Fig. 1.3.

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- (ii) Using the information provided and Fig. 1.3, explain how Gramicidin A kills the bacterium.

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- (f) Upon prescription of antibiotics, doctors often advise patients to complete the course of antibiotics even if symptoms of disease have ceased. One of the reasons cited was that not completing the course of antibiotics may increase the chance of antibiotic resistant bacteria to evolve.

With reference to natural selection, explain the basis for the need to complete the prescribed course of antibiotic.

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..... [3]

[Total: 25]

- 2 A researcher investigated the pathway by which carbon dioxide is converted to organic compounds during photosynthesis. The apparatus used is shown in Fig. 2.1.

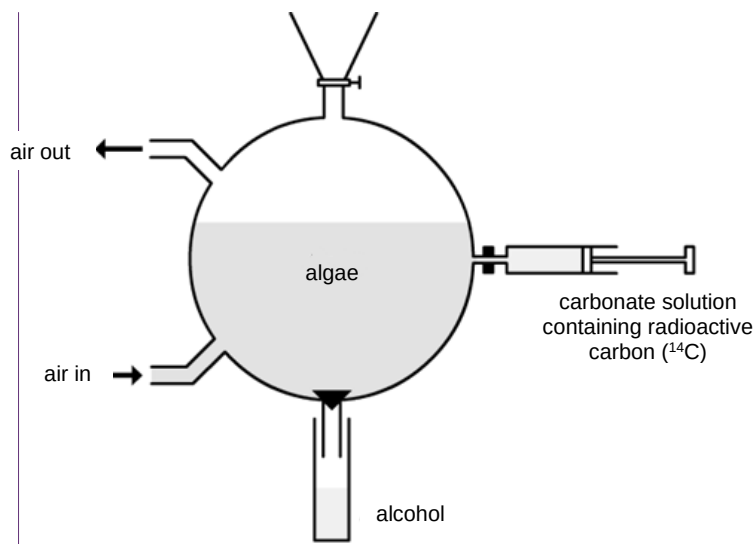


Fig. 2.1

While the apparatus was in the dark room, the researcher supplied the algal cells with $^{14}\text{CO}_2$. The contents of the apparatus were thoroughly mixed and light was switched on subsequently. At five-second intervals, a few of the cells were released into hot alcohol, which killed the cells very quickly. The intermediates of the reactions were subsequently analysed and the chromatograms in Fig. 2.2 showed the results of the analysed intermediates by chromatography (a separation technique) at different timings.

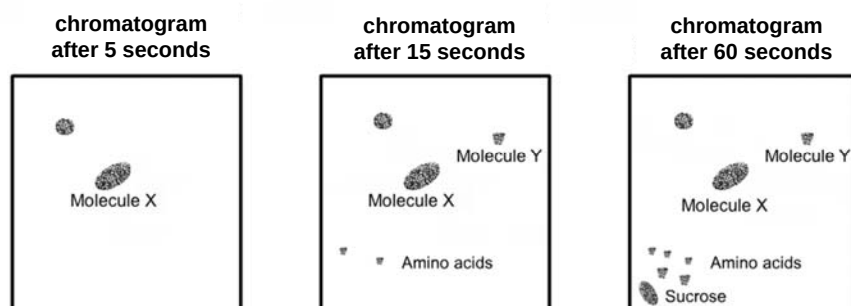


Fig. 2.2

- (a) (i) Identify molecules X and Y.

X

Y

[2]

Commented [A1]: need to have an "air out" textbox? ok

Commented [A2]: Need to explain what chromatogram is? Are students expected to know chromatographic separation? This similar question also appeared in our PS tutorial. Also chromatogram is taught in Chem?

- (ii) Explain your answer in part (i).

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[4]

- (iii) Explain why sucrose and amino acids are identified in the chromatogram only after 60 seconds.

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[1]

Dinitrophenol is a metabolic poison that can embed within the thylakoid membranes of chloroplasts and provide an alternate route for H^+ to diffuse across the thylakoid membranes.

- (b) Explain how the concentration of intermediates of the Calvin cycle is affected by dinitrophenol.

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[3]

Commented [A3]: To give then idea that these molecules are synthesized later? So with the word only here will suffice?

Another experiment was carried out by another student to determine the concentration of carbon dioxide in the leaves of plants at different times of the day. The results are shown in Table 2.3

Table 2.3

Mean carbon dioxide concentration (ppm)	
8pm to 4am	8am to 4pm
328	106

- (c) Using knowledge on Calvin cycle and Krebs cycle, account for the difference in concentration of carbon dioxide in the leaves for the two periods shown in Table 2.3.

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- (d) Explain the effect of higher concentration of carbon dioxide on the rate of carbon fixation during the period 8am to 4pm.

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Studies were carried out on soil-dwelling aerobic and anaerobic bacteria. Samples were taken from different depths at intervals of one month and six months after the soil was put into a large heap for storage.

Table 2.4 shows the numbers of aerobic and anaerobic bacteria at different depths in the stored soil.

Table 2.4

depth in soil store / m	mean number of bacteria per gram of stored soil $\times 10^7$			
	aerobic bacteria		anaerobic bacteria	
	after one month	after six months	after one month	after six months
0.0	12.4	12.5	0.4	0.6
0.5	10.1	8.3	0.6	1.0
1.0	9.8	5.9	0.8	3.8
1.5	9.7	3.1	0.8	7.6
2.0	10.5	0.8	0.7	8.1
2.5	10.8	0.7	0.8	8.5
3.0	10.2	0.9	0.6	8.8

- (e) (i) Account for the trends shown by the distribution of the two types of bacteria after six months.

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..... [3]

- (ii) Suggest how aerobic bacteria are structurally adapted for cellular respiration.

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In a further study, soil samples were taken at two depths, **A** and **B**, in the soil store. The samples were taken at intervals over six years. Soil samples of equal mass were used to determine the activity of dehydrogenases in aerobic bacteria.

Fig. 2.5 shows the mean dehydrogenase activity of the bacteria in these samples.

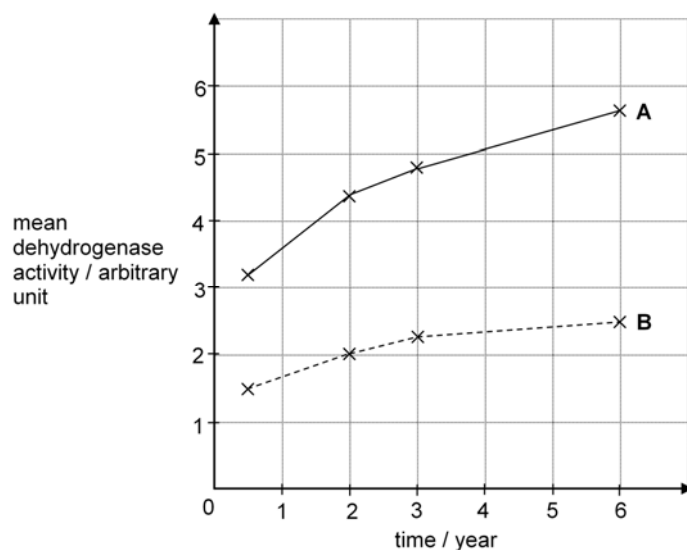


Fig. 2.5

- (f) (i) State with evidence from Fig. 2.5 which depth, **A** or **B**, were samples taken from a greater depth.

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..... [2]

- (ii) Explain the roles of dehydrogenase in Krebs cycle of the aerobic bacteria.

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..... [2]

[Total: 25]

Commented [A4]: Students may be confused as bacteria do not have mito. They may wonder why Krebs cycle is still possible. So remove 'Krebs cycle'?

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts (a), (b), etc., as indicated in the question.

- 3 (a)** Greenhouse gases are key contributors to climate change affecting animals and plants in the environment they live in.

Discuss the effects of climate change in the global environment. [13]

- (b)** After viruses infect host organisms, they are able to make use of host cell machinery to replicate and reproduce thereby causing diseases in the host organism.

Describe how dengue causes viral disease in humans. [12]

[Total: 25]

- 4 (a)** Effector molecules are responsible for the regulation of transcriptional units in prokaryotes.

Using named examples, explain the roles of these effector molecules in the negative feedback regulation of transcriptional unit in a prokaryote such as *Escherichia coli*. [12]

- (b)** Protein production in eukaryotes is controlled at all stages of the process.

Explain how protein production is controlled in eukaryotes and the advantages of regulating protein production at different stages [13]

[Total: 25]

For
Examiner's
Use

Commented [A5]: what is the expected answer? the pathology of infection ? Yes. Perhaps change to "Describe how dengue causes viral disease in human"

Commented [A6]: Should we state "using a named example"? Ok.

Commented [A7]: do the students know about the advantage at each stage? This is a question from specimen paper. Hence, need to let them know.

[WMY] The MS only requires advantage from any 4 stages. Kind of misleading hor? Yes it is. Still can right?

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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